1. System Overview & Core Architectural Pillars

How Sift transforms high-volume unstructured feedback into verified roadmap action items

THE B2B SAAS FEEDBACK CRISIS & SIFT'S APPROACH

Modern product teams face thousands of unstructured feedback rows across Zendesk, Gong, Intercom, and G2. Traditional tools create 'Unweighted Volume Illusions' (10 free users requesting dark mode outvote 1 Enterprise account threatening to churn over a data export timeout). LLM chatbots invent hallucinated quotes and ungrounded tech specs. Sift couples client-side deterministic math with controlled 2-pass AI clustering and verbatim substring verification.

Pillar 1: Ephemeral Local-First Privacy

- 100% in-browser CSV parsing via PapaParse (no database).
- Stateless serverless proxy with ZERO payload logging.
- API keys locked on edge; client bundles contain zero secrets.
- Unprocessed rows remain in ephemeral browser state.

Pillar 2: Deterministic Hybrid Math (No LLM Guesswork)

- Score: $P_c = \text{Round}_1(100 \times [W_{ARR} \cdot I_n + W_{Vol} \cdot F_n + W_{Sev} \cdot S_n])$.
- Live sliders recalculate rank in <1.5 ms (60 FPS benchmark).
- Stationary Coordinates: Matrix dots (F_n, I_n) NEVER move on drag.
- UNKNOWN tiers isolated at 1.0x baseline weight multiplier.

Pillar 3: Verbatim Quote Verification (IC-001)

- Exact substring inclusion against source CSV row in memory.
- Zero .trim(), zero lowercasing, zero punctuation normalization.
- Exact matches get green 'Verified Quote' badge.
- Normalized near-matches explicitly flagged 'Not Verbatim Verified'.

Pillar 4: 5-Tier Safety Decoupling (PRD §7.4)

- Raw observations separated from AI root-cause hypotheses.
- Missing tech context triggers '[NEEDS INVESTIGATION]' tags.
- Copy-ready PM Strategic Rationales & Dev Engineering Specs.
- Zero AI hallucination of databases, APIs, or architectural stacks.

END-TO-END DISCOVERY & TRIAGE PIPELINE

1. Ingestion & Schema

Drop 300–2,000 row CSV
Auto-detect 7 columns
Isolate UNKNOWN tiers

→

2. Two-Pass AI Proxy

50-row batching (~8k tok)
Pass 1 cluster formulation
Pass 2 category reconcile

→

3. Math & 2x2 Matrix

Deterministic P_c scoring
4 quadrant categorization
60 FPS live slider ranks

→

4. Evidence & Specs

IC-001 quote verification
[NEEDS INVESTIGATION]
Copy PM & Eng deliverables

2. CSV Blueprint & 7 Canonical Columns Specification

Standardize messy feedback exports while guaranteeing zero data loss and strict privacy

INGESTION ENGINE PARAMETERS & DATA LIMITS (REQ-001 & REQ-002)

- Parsing: Browser-local PapaParse with auto delimiter detection (comma, tab, semicolon) and UTF-8/UTF-16 encoding.
- Capacity Ceiling: Up to 10 MB or 2,000 rows parsed in memory. (300 rows benchmarks in <40 ms; ceiling benchmarks in <0.5 s).
- Privacy Invariant: Raw feedback text never leaves client memory except via chunked JSON to the stateless proxy route.

CANONICAL 7-COLUMN SCHEMA MAPPING GRID

Column Name	Key	Req	Recognized Aliases / Headers	Handling, Weight Multipliers & Fallback Rules
Raw Feedback Text	`text`	**Required**	feedback, text, comment, description, notes, review, issue_description	Mandatory core text. Preserved verbatim. Missing column blocks ingestion with explicit error toast.
Customer Tier / ARR	`tier`	**Optional**	tier, plan, arr, mrr, customer_tier, account_type, segment	Mapped to Enterprise (3.0x), Pro (2.0x), Free (1.0x). Missing/invalid rows set to UNKNOWN (1.0x proxy).
Feedback ID	`feedback_id`	**Optional**	id, uuid, row_id, ticket_id, feedback_id, reference_id	Preserved verbatim. If omitted, Sift assigns stable sequential IDs (FB-001). Duplicates get '_row2' suffix.
Account ID	`account_id`	**Optional**	account_id, company, org_id, domain, customer_name, organization	Paying corporate entity. Dual-reporting: Confirmed accounts vs Estimated rows. Triggers warning pill if unmapped.
User ID / Submitter	`user_id`	**Optional**	user_id, author_id, email, submitter, user_email, contact_id	Individual human submitter. Enables user-level incident counts while strictly decoupled from corporate accounts.
Source Channel	`source`	**Optional**	channel, source, touchpoint, platform, origin, intake_method	Categorizes origin (Zendesk, Gong, Intercom, G2, In-App). Preserved in deep-dive evidence panel.
Date Submitted	`date`	**Optional**	date, created_at, timestamp, time, submitted_at, log_date	Temporal tracking for churn velocity and recency. Preserved in raw metadata inspection table.

CRITICAL DATA QUALITY & IDENTITY GUARDRAILS

Data Quality Warning Banner (IC-012)

Displayed whenever UNKNOWN tier or unmapped accounts >5%. Notice: 'Data Quality Warning: N% of records lack tier/account metadata and are evaluated provisionally (1.0x weight). Scores may reflect volume over ARR.'

Dual Account Identity Counts (REQ-003)

Sift never fabricates corporate names. Dual reporting:
- Confirmed: Deduplicated unique Account IDs.
- Estimated: 1 row = 1 provisional account (U_c = N_c).
- Warning pill: 'Account IDs Unmapped — Frequency Estimated'.

Strict UNKNOWN Tier Isolation (DEC-003)

Missing customer tier is NEVER auto-coerced to Free or $0. Carries a neutral 1.0x weight multiplier (preventing score distortion) while maintaining clear visual badges in deep-dive drawer.

Verbatim Cell Storage (IC-001)

PapaParse outputs are stored without trimming, case conversion, or character stripping. Exact substrings guarantee downstream receipt verification passes without false negative mismatches.

3. 2x2 Strategic Priority Matrix & 60-FPS Dynamic Sliders

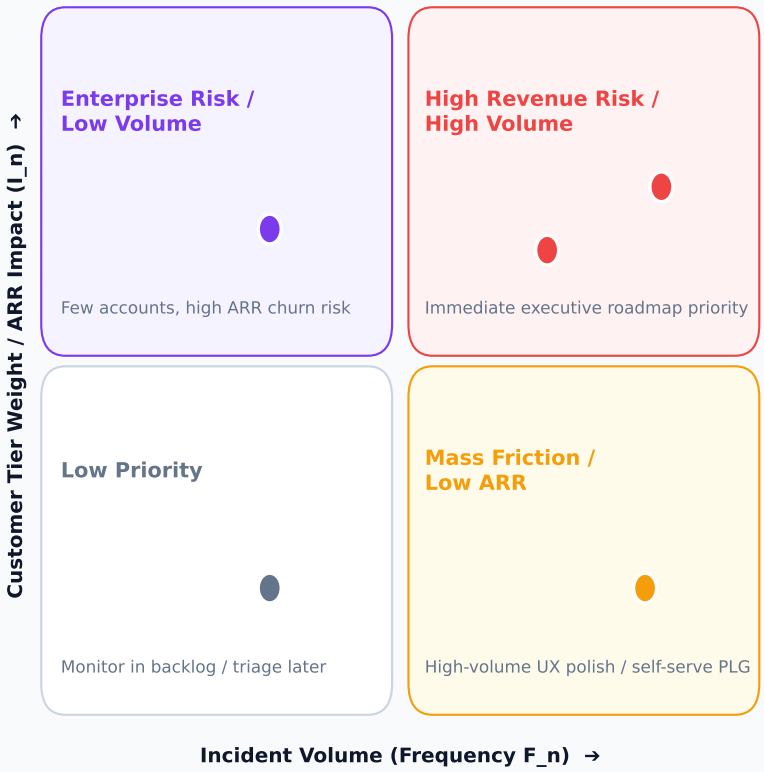
Visualizing ARR revenue risk vs incident volume with stationary physical coordinate invariants

HYBRID DETERMINISTIC SCORING FORMULA (PRD §6.1 & MDD §3.5)

$P_c = \text{Round}_1(100 \times [W_{ARR} \cdot I_n + W_{Vol} \cdot F_n + W_{Sev} \cdot S_n])$ | Defaults: 40% ARR Impact, 40% Volume, 20% Severity

- Normalized Revenue Impact (I_n): Weighted tier sum divided by max dataset cluster sum.
- Normalized Volume (F_n): Log-normalized frequency count (F_n = ln(1 + N_c) / ln(1 + N_max)).
- Severity Score (S_n): Direct mapping from cluster severity (Blocker=1.0, Major=0.6, Minor=0.3).

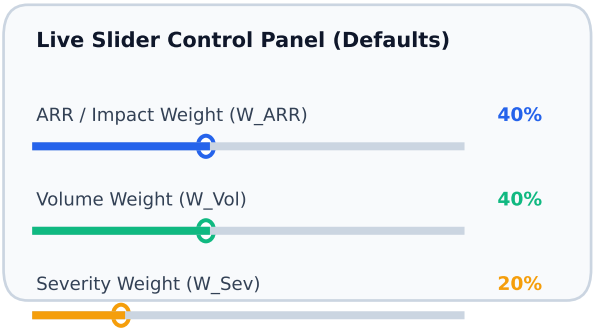
THE 2x2 STRATEGIC DECISION MATRIX



STATIONARY COORDINATE INVARIANT (DEC-010)

CRITICAL PRODUCT RULE (MDD §4.2 & TS-006):
When a user drags weighting sliders (ARR, Volume, Severity), matrix dots (F_n, I_n) DO NOT MOVE on the chart.

- Why? Physical reality does not change when internal team priorities change. Volume is volume; tier impact is tier impact.
- What updates? Cluster Priority Score (P_c), quadrant badge, and the sorted card list in the adjacent view.
- Latency: Under 1.5 ms execution benchmark ensures buttery-smooth 60 FPS responsiveness even with 1,000+ clusters.



CANONICAL 2x2 QUADRANT POLICY & ACTION BLUEPRINT (IC-014)

High Revenue Risk / High Volume

Top-Right: Chronic blockers impacting enterprise ARR.
Action: Sprint commitment; generate Engineering Spec.

Enterprise Risk / Low Volume

Top-Left: Bespoke enterprise pain points & churn threats.
Action: CS advisory; executive discovery call.

Mass Friction / Low ARR

Bottom-Right: Broad UX paper cuts across self-serve tiers.
Action: PLG backlog; batch into UI polish sprints.

Low Priority

Bottom-Left: Niche or cosmetic feedback with minimal impact.
Action: Retain in backlog; monitor future cohort trend.

4. Deep-Dive Drawer, Quote Verification & Spec Generation

Auditing exact evidence receipts and generating copy-ready deliverables for PMs and Engineers

THE DEEP-DIVE EVIDENCE DRAWER

Clicking any cluster card opens an accessible slide-over drawer exposing four decoupled intelligence layers:

1. Metric Bar: Priority Score (P_c), Severity Badge (1–3), Affected Accounts count (Confirmed vs Estimated).
2. AI Root-Cause Hypothesis: Demarcated as 'AI-Drafted', explaining mechanical failure (e.g. 60s gateway timeout).
3. Traceable Evidence Panel: Verbatim quote receipts, exact substring verification badges, and source row reveal.
4. Action Output Hub: One-click export of copy-ready PM Strategic Rationales and Developer Engineering Specs.

EXACT QUOTE VERIFICATION (IC-001 & TS-005)

SIFT'S CARDINAL VERIFICATION RULE:
Quotes displayed in evidence panels MUST BE exact verbatim substrings of the source CSV row. No paraphrasing, no rewording.

Badges & Guardrails:
- [Verified Quote] (Green Badge): Exact case-sensitive match.
- [Near Match / Normalized — Not Verbatim Verified] (Amber): Displayed if punctuation/whitespace was modified by LLM.
- [Source row not found]: Displayed if Feedback ID is missing.
- [Reveal Full Raw Row]: Interactive toggle revealing unmutated raw text directly from the parsed CSV in memory.

THE 5-TIER SAFETY DECOUPLING ARCHITECTURE (DEC-005 & TS-013)

Tier 1: Observation	Tier 2: Verified Quotes	Tier 3: AI Hypothesis	Tier 4: Investigations	Tier 5: Eng Notes
Raw user feedback preserved in client memory	IC-001 exact substring receipts only	Thematic failure root-cause labeled 'AI-Drafted'	Injected [NEEDS INVESTIGATION] placeholders for missing context	Human architecture decisions decoupled from AI guessing

COPY-READY ROLE-TAILORED DELIVERABLES GENERATOR

PM Strategic Trade-Off Rationale (PRD §7.4)

Executive decision memo format with 5 canonical sections:
- Header: Priority Score (P_c), Severity, Confirmed/Estimated accounts.
- Section 1: Executive Summary & Customer Problem Statement.
- Section 2: Business & ARR Impact (strict proxy tier weights; never labels proxy scores as actual dollars per IC-002).
- Section 3: Verbatim Customer Evidence Receipts (verified quotes).
- Section 4: Strategic Recommendation (Suggested vs AI-Drafted).
- Section 5: Contradictory Evidence Boundary (GAP-07 / P1 marker).

Developer Engineering Spec (PRD §7.4 & TS-013)

Copy-ready technical implementation ticket for engineering:
- Problem Statement: Thematic cluster and root-cause hypothesis.
- Reproduction Steps: Derived from source feedback rows.
- Affected Customer Accounts: Deduplicated confirmed account list.
- Safety Placeholders (DEC-005): When browser/OS or stack context is missing, automatically injects: '[NEEDS INVESTIGATION: Technical Context Missing in Source Feedback]'
- Clipboard Reliability: Auto-copies with textarea fallback.